

Attack-Free Early Warning and Timely PGD Escalation for Catastrophic Overfitting

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Abstract—Single-step adversarial training is efficient but can abruptly lose robustness through catastrophic overfitting (CO). We study an online safety trigger based on the cosine consistency of two input gradients obtained within an FGSM parameter update. The CIFAR-10 threshold is selected on seeds 17, 23, and 42 and frozen before held-out evaluation. It detects 7/7 held-out CIFAR-10/PreActResNet-18 events within 60 updates. With the threshold unchanged, it detects 4/5 CIFAR-10/ResNet-18 events, including 2/5 within 60 updates. After one dataset-specific calibration, it detects both observed held-out CIFAR-100 events; the three CIFAR-100 non-event trajectories contain one alarm episode. Eight randomized-FGSM runs at $\epsilon = 8/255$ remain stable and produce no alarms. At $\epsilon = 16/255$, all three stress-test runs naturally collapse; the frozen CIFAR-10 detector warns 20–60 updates beforehand, and the tested *immediate-replay* ablation raises terminal PGD-10 accuracy from 0 to $26.48 \pm 0.83\%$. In a separate five-seed timing audit, both *immediate-switch* and the rollback ablation *immediate-replay* start PGD-2 before CO in 5/5 cases, versus 3/5 for next-epoch escalation. For *immediate-switch*, final PGD-10 accuracy is $47.05 \pm 0.74\%$; its +0.22-point difference from *immediate-replay* has a paired 95% interval crossing zero. Immediate switching provides the timing benefit; rollback adds no reliable endpoint gain. In controlled reimplementations, GradAlign achieves higher robustness, while the proposed policy uses lower measured core training time. For the fixed seed-101 *immediate-switch* checkpoint, full-test PGD-50, five-restart PGD-20, and fixed-subset AutoAttack-standard accuracy are 45.52%, 45.66%, and 42.30%, respectively. Gradient consistency is used as an empirical warning signal; conditional PGD-2 supplies the recovery.

I. INTRODUCTION

Deep classifiers can change their predictions under small adversarial perturbations [1], [2]. Adversarial training addresses this failure by optimizing against perturbed examples and remains a strong empirical defense [3]. Its main obstacle is cost because multi-step training attacks require several forward-backward passes for nearly every parameter update.

Fast adversarial training replaces the inner loop with a single FGSM step. With randomized initialization and a suitable schedule, it can approach the robustness of PGD training at much lower cost [4]. Yet single-step training can exhibit *catastrophic overfitting* (CO): iterative-attack accuracy falls sharply toward zero while clean or FGSM accuracy remains high. FastAdv+ responds using periodic PGD validation [5]; GradAlign regularizes local input-gradient geometry at every update [6]. These approaches improve reliability but add attack or gradient computation.

We ask whether the FGSM update itself provides an early warning. It computes an input gradient at the attack starting point, and the endpoint input gradient is obtained from the same parameter-backward pass without an additional forward or backward pass. Their cosine similarity therefore requires no additional model evaluation. We use “attack-free monitor” to mean that the monitor itself does not run an auxiliary validation attack. The base training still generates FGSM adversarial examples, and validation PGD-10 is still used to select *best_robust.pt*. When the cosine-drop score crosses a frozen threshold, the primary *immediate-switch* response retains the completed update, switches the next mini-batch to PGD-2, and continues with PGD-2. The matched *immediate-replay* ablation instead rolls back to the epoch-start state before recovery.

The contributions are:

- A gradient-consistency safety trigger calibrated on three CIFAR-10 development seeds and audited on seven disjoint held-out events. We also test unchanged-threshold architecture transfer and protocol transfer after one CIFAR-100-specific calibration.
- An event-level timing study that records warning, unprotected CO, and actual PGD-2 start times. Immediate-switch and immediate-replay act before collapse in 5/5 matched events, compared with 3/5 for next-epoch escalation; five seeds do not establish a clear proportional advantage.
- Five-seed component controls and controlled FastAdv+/GradAlign reimplementations under one harness, with paired bootstrap intervals.
- A random-start stress test: the monitor stays silent on eight stable 8/255 runs but detects 3/3 natural CO events at 16/255; the tested immediate-replay ablation restores non-trivial robustness.

The study concerns empirical ℓ_∞ robustness. Certified defenses provide formal guarantees through different mechanisms and are complementary to the training-time monitor studied here.

II. RELATED WORK

A. Adversarial Attacks and Training

FGSM follows the sign of the input loss gradient in one step [2]. PGD iterates gradient steps with projection onto a prescribed threat set and is central to the robust-optimization view of adversarial training [3]. TRADES separates natural error from a boundary-related robustness term and makes the

clean-robust trade-off explicit [7]. Although these methods are effective, their multi-step inner loops increase training cost. Free adversarial training reuses gradients across repeated mini-batches [8], whereas fast training uses a single randomized FGSM step [4].

Robustness evaluation itself must be sufficiently strong. PGD results can be sensitive to iterations, restarts, and objectives. AutoAttack combines complementary parameter-free attacks to reduce evaluator bias and expose gradient masking [9]. In the controlled study below, we use progressively stronger full-test PGD variants and multiple restarts rather than treating FGSM accuracy as a robustness estimate.

B. Catastrophic Overfitting

Wong *et al.* identified CO as a failure mode of fast adversarial training [4]. Andriushchenko and Flammarion connected it to local nonlinearity and proposed GradAlign, which regularizes gradient alignment inside the perturbation set [6]. Li *et al.* showed that recovery from weak-attack overfitting is important and proposed FastAdv+ and FastAdvW, which use periodic PGD validation to decide when to introduce PGD updates [5]. Our response mechanism is related in spirit, but the detector differs operationally: the endpoint input gradient is obtained from the same parameter-backward pass, without an additional forward or backward pass. The monitor does not execute a validation attack. We also separate prediction from mitigation by freezing the detector before held-out evaluation.

C. Empirical and Certified Robustness

Empirical defenses are evaluated against known attacks but do not guarantee correctness for every allowed perturbation. Certified methods instead compute a provable lower bound, for example through convex outer approximations [10] or randomized smoothing under an ℓ_2 threat model [11]. This paper targets efficient empirical training in the ℓ_∞ setting. The distinction is important: the attack results reported below measure resistance to a strong test suite, not certified radii.

III. METHOD

A. Threat Model and Fast Update

Let f_θ be a classifier and ℓ cross-entropy loss. Adversarial training under an ℓ_∞ radius ϵ approximates

$$\min_{\theta} \mathbb{E}_{(x,y)} \left[\max_{\|\delta\|_\infty \leq \epsilon} \ell(f_\theta(x + \delta), y) \right]. \quad (1)$$

FGSM begins at x^0 , either x or a random point in the perturbation set, and constructs

$$g^0 = \nabla_{x^0} \ell(f_\theta(x^0), y), \quad x^{\text{adv}} = \Pi(x^0 + \alpha \text{sign}(g^0)), \quad (2)$$

where Π projects onto the valid image and threat set.

B. Attack-Free Gradient-Consistency Score

The endpoint input gradient $g^1 = \nabla_{x^{\text{adv}}} \ell(f_\theta(x^{\text{adv}}), y)$ is obtained from the same parameter-backward pass, without an additional forward or backward pass. We retain it and calculate

$$c(x) = \frac{\langle g^0, g^1 \rangle}{\|g^0\|_2 \|g^1\|_2}. \quad (3)$$

FGSM relies on a local first-order approximation. A sharp decrease in the cosine between g^0 and g^1 means that the input gradient rotates along the FGSM direction, so the local linear approximation and the representativeness of the single-step inner maximizer are degrading. This interpretation agrees with prior links between local nonlinearity and CO [6]. We do not claim that the cosine drop is a necessary or sufficient condition for CO; it is an empirically validated proxy for the local-linearity failure associated with CO.

Batch averages are accumulated over non-overlapping windows of 20 optimizer updates to give c_t . The monitor adds reductions and trace state, but no additional forward or backward pass. In the implementation, g^0 constructs the FGSM example and g^1 is retained from the same parameter-backward pass; the monitor calls no additional autograd-gradient operation. To remove the slow training trend, it uses

$$b_t = \text{median}(c_{t-5}, \dots, c_{t-1}), \quad s_t = b_t - c_t, \quad (4)$$

after at least three history windows. A warning occurs when $s_t \geq \tau$. For CIFAR-10 calibration, we pool every finite development score, label a window positive only when a registered CO event is the next 20-update window, scan the unique observed scores, and choose the threshold maximizing Youden's J (true-positive rate minus false-positive rate). Seeds 17, 23, and 42 form the development set; this yields $\tau = 0.1663298875$, which is then frozen. CIFAR-100 uses a separate event-level lexicographic rule, detailed in Section IV. Offline sensitivity analysis in the Appendix does not alter either fitted value or the main protocol.

C. Immediate-Switch Conditional Escalation

At the first warning, the primary response retains the completed FGSM update, switches the next mini-batch to PGD-2, and uses PGD-2 for all subsequent updates. Thus no additional FGSM update intervenes between warning and action, and no training state must be restored.

We call this no-replay main response *immediate-switch*. Four matched alternatives are evaluated. The rollback ablation, *immediate-replay*, keeps an in-memory epoch-start snapshot of model, optimizer, mixed-precision scaler, Python, NumPy, PyTorch CPU/CUDA, and loader-generator random states. At the warning it restores that state and replays the epoch using PGD-2; the snapshot is not serialized to disk. The remaining controls retain the current FGSM epoch and start PGD-2 at the next epoch, switch at a fixed early or late epoch, or use PGD-2 from epoch 1. These controls separate the value of prompt action from any effect of restoring the epoch.

D. Event and Warning/Evaluation Protocol

On fixed validation data, a CO event requires PGD-10 accuracy to fall at least 20 percentage points from its previous peak and reach at most 10%. We report whether a warning precedes the first event, whether it occurs within 60 updates, median lead, and alarm episodes in non-event runs. Validation PGD is used for retrospective event labels and for selecting *best_robust.pt*; it never triggers the proposed monitor. Thus “attack-free” describes the online monitor, not an attack-free checkpoint-selection protocol. Recall intervals are Wilson 95% intervals. Accuracy and time comparisons use identical seeds and paired bootstrap 95% intervals with 100,000 resamples.

IV. EXPERIMENTS

A. Setup and Seed Roles

The primary setting is CIFAR-10 with PreActResNet-18 [12], [13], a deterministic 45,000/5,000 training/validation split, and $\epsilon = 8/255$. All methods use SGD with momentum 0.9, weight decay 5×10^{-4} , batch size 128, mixed precision, 30 epochs, and a triangular learning-rate schedule. The collapse-prone recipe uses zero-start FGSM, step 8/255, and peak learning rate 0.3. The ordinary FGSM-RS control uses a uniform random start, step 10/255, and peak learning rate 0.2. Recovery uses PGD-2 with step 4/255.

We keep experimental roles explicit. Detector development uses CIFAR-10 seeds 17, 23, and 42. Held-out detector evaluation uses seeds 101, 202, 303, 404, 505, 606, and 707. Intervention ablations and baseline comparisons use the matched set 17, 23, 42, 101, and 202. Non-event controls are reported separately from event recall. Seeds 17, 23, and 42 are used for threshold calibration. They appear in intervention ablations but are not counted toward held-out detector-evaluation recall.

Transfer tests change one factor at a time. CIFAR-10/ResNet-18 uses the unchanged CIFAR-10 threshold on event seeds 303–707 and three non-event runs at seeds 17, 23, and 42. CIFAR-100 uses PreActResNet-18 and the same 20-update windows, five-window median, and three-window minimum history. Saved development traces at seeds 17, 23, and 42 set a one-time dataset-specific threshold of 0.6332649834303036 before evaluation of held-out event candidates 101, 202, and 303 and two random-start non-events. A separate CIFAR-10 threat-budget stress test retains random starts and uses $\epsilon = 16/255$, FGSM step 20/255, and peak learning rate 0.2.

The CIFAR-100 calibration uses six development trajectories: one collapse-prone and one random-start run for each of seeds 17, 23, and 42. Its original fitting code scans unique finite scores and applies a lexicographic event-level rule: maximize detected development events, minimize alarm episodes on stable development runs, prefer shorter median first-warning lead, and then prefer the larger threshold. Only one of the three collapse-prone development runs meets the registered CO definition. The fitted threshold is 0.6332649834303036. This dataset-specific procedure differs from the CIFAR-10 window-level Youden- J calibration.

B. Held-Out Detection and Transfer

The primary held-out CIFAR-10 evaluation contains seven registered events spanning epochs 13–24, and none uses a threshold-development seed. The frozen monitor is active within 60 updates before all 7/7 events; the median lead of the earliest active warning inside that 60-update window is 40 updates. One event trajectory also contains an earlier alarm episode. The three ordinary FGSM-RS non-event runs contain no alarm episode. On the same event traces, a periodic PGD-10 probe every 100 updates detects 6/7 events with a median delay of 40 updates after the event and makes a median of 73 PGD calls before failure. The 100-update probe is a retrospective detector comparator applied to the saved collapse trajectories; it does not alter training. By contrast, the controlled FastAdv+ reimplementation probes a fixed 128-image validation batch every 20 training updates and uses the observed accuracy drop to trigger a transient PGD training block. The two frequencies therefore describe different protocols and computational budgets.

Table I separates observed events from all non-events. With the CIFAR-10 threshold unchanged, ResNet-18 yields 4/5 detected events but only 2/5 within 60 updates; this is partial unchanged-threshold architecture transfer, not complete cross-architecture generalization. On CIFAR-100, two of three planned collapse-prone trajectories meet the event definition and both are detected after the one-time dataset-specific calibration. The third collapse-prone trajectory does not meet the registered CO definition and is therefore counted with the two random-start controls: among these three non-events there is one alarm episode.

C. Randomized-FGSM Stress Test

We first test eight randomized-FGSM runs at $\epsilon = 8/255$: three standard runs at peak learning rate 0.2 and five runs at 0.3. All eight remain robust, and the detector produces zero alarm episodes. The monitor does not intervene in this tested standard 8/255 configuration.

Prior work reports that random starts become less reliable at larger budgets [6]. At $\epsilon = 16/255$, all three random-start runs meet the CO definition and terminate at 0% PGD-10. The frozen CIFAR-10 detector warns 20, 40, and 60 updates before collapse. The intervention logs identify these three runs as the immediate-replay ablation, not the immediate-switch main method. Immediate-replay begins PGD-2 before every event and finishes at 25.55%, 26.80%, and 27.11% PGD-10. Mean clean and PGD-10 accuracy are $71.05 \pm 0.43\%$ and $26.48 \pm 0.83\%$. This is a threat-budget stress test, not cross-dataset evidence.

D. Actionable Lead and Component Controls

Table II distinguishes statistical lead from usable lead. All three responses receive the identical first warning. Immediate-switch and immediate-replay begin PGD-2 with no intervening FGSM update and act before all five matched collapses. Waiting for the epoch boundary adds a median of 212 FGSM updates; two collapses occur before PGD-2 starts.

TABLE I

DETECTOR AUDIT. “NEAR” DENOTES AN ACTIVE WARNING WITHIN 60 UPDATES. THE LAST COLUMN IS THE MEDIAN NEAR-WARNING LEAD IN UPDATES.

Dataset / budget	Architecture	Planned	Events	Detected	Near	Non-event runs / alarm episodes	Median near lead
CIFAR-10, 8/255	PreActResNet-18	7	7	7/7	7/7	3 / 0	40
CIFAR-10, 8/255	ResNet-18	5	5	4/5	2/5	3 / 0	80
CIFAR-100, 8/255	PreActResNet-18	3	2	2/2	2/2	3 / 1	40
CIFAR-10, 16/255 RS	PreActResNet-18	3	3	3/3	3/3	–	40

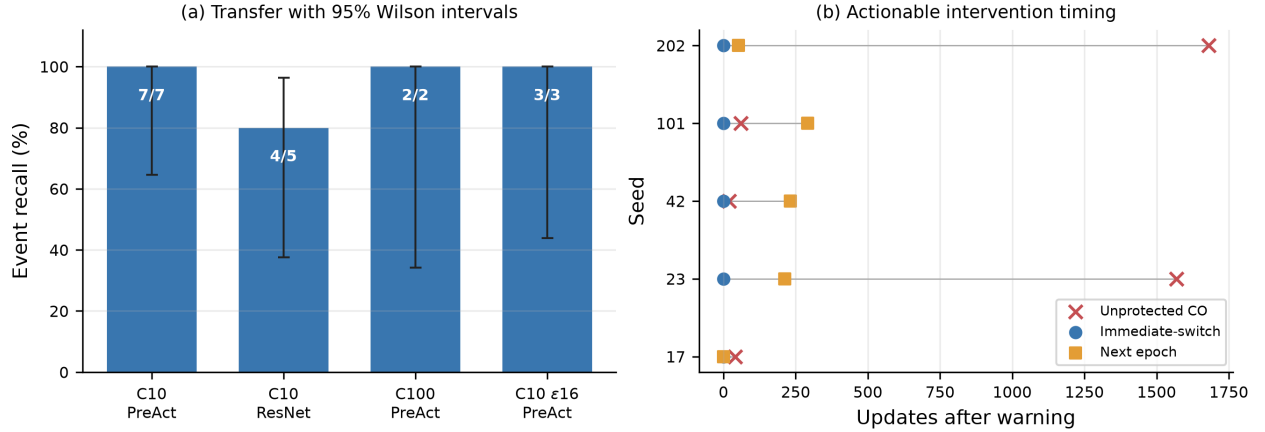


Fig. 1. Left: event recall across the primary setting, unchanged-threshold architecture transfer, dataset-calibrated transfer, and a threat-budget stress test, with Wilson 95% intervals. Right: immediate-switch and immediate-replay start PGD-2 with zero intervening FGSM updates; next-epoch escalation starts after unprotected CO in two of five matched trajectories.

TABLE II

FIVE-SEED WARNING-TO-ACTION AUDIT (MEAN $\pm$ SAMPLE STD.). WARN $\rightarrow$ START COUNTS INTERVENING FGSM OPTIMIZER UPDATES.

Response	Before CO	Warn $\rightarrow$ start	Clean	PGD-10	Core min
Immediate-switch (ours)	5/5	0 updates	83.06 $\pm$ 0.65	47.05 $\pm$ 0.74	18.74 $\pm$ 0.63
Immediate-replay (ablation)	5/5	0 updates	83.14 $\pm$ 0.62	46.83 $\pm$ 0.55	18.68 $\pm$ 0.70
Next epoch	3/5	212 updates	83.09 $\pm$ 0.57	46.72 $\pm$ 1.05	18.14 $\pm$ 0.58

All component controls use seeds 17, 23, 42, 101, and 202. Immediate-switch differs from immediate-replay by +0.22 PGD-10 points (paired bootstrap 95% CI $[-0.22, +0.66]$) and +0.06 core minutes (CI $[-0.18, +0.30]$); the clean difference is -0.07 points (CI $[-0.16, 0.00]$). Neither comparison supports an immediate-replay advantage. Relative to waiting for the next epoch, immediate-switch starts PGD-2 157.6 updates earlier on average (CI $[-252.0, -56.8]$), covers 5/5 rather than 3/5 events before CO, with a proportion difference of +0.40 (CI $[0.00, +0.80]$), and differs by +0.33 PGD-10 points (CI $[+0.08, +0.66]$). Its clean difference is -0.02 points (CI $[-0.14, +0.08]$), while core time increases by 0.59 minutes (CI $[+0.47, +0.71]$). The before-CO point estimate favors immediate-switch, but five seeds are insufficient to establish a clear proportional advantage. The small endpoint difference is secondary to the intervention-timing result. Full PGD-2 differs from the next-epoch policy by +0.09 points (CI $[-0.27, +0.42]$) but adds 3.46 core minutes. The detector supplies the warning, PGD-2 supplies recovery, and immediate-switch converts warning into pre-collapse action; rollback itself is not the active ingredient.

TABLE III

MATCHED FIVE-SEED CONTROLS (MEAN $\pm$ SAMPLE STD.)

Policy	Clean	PGD-10	Core min
Immediate-switch (ours)	83.06 $\pm$ 0.65	47.05 $\pm$ 0.74	18.74 $\pm$ 0.63
Immediate-replay (ablation)	83.14 $\pm$ 0.62	46.83 $\pm$ 0.55	18.68 $\pm$ 0.70
Next epoch	83.09 $\pm$ 0.57	46.72 $\pm$ 1.05	18.14 $\pm$ 0.58
Fixed epoch 13	83.32 $\pm$ 0.70	46.41 $\pm$ 0.50	19.37 $\pm$ 0.21
Fixed epoch 21	82.35 $\pm$ 1.12	45.75 $\pm$ 0.94	17.85 $\pm$ 0.21
PGD-2 from epoch 1	83.57 $\pm$ 1.02	46.81 $\pm$ 0.90	21.60 $\pm$ 0.13

E. Controlled CO-Mitigation Baselines

FastAdv+ and GradAlign are controlled reimplementations rather than claims of using official released code. They share the split, architecture, five seeds, 30-epoch schedule, checkpoint rule, and PGD-10 evaluator. FastAdv+ uses one fixed 128-image validation batch, a PGD-10 probe every 20 optimizer updates with step 2/255, a 0.10 drop from the best observed probe accuracy as its trigger, and a transient 20-update PGD-10 training block before returning to random-start FGSM. GradAlign uses $\lambda = 0.2$, a uniform perturbation in $[-\epsilon, \epsilon]$ clipped to the image domain, cosine alignment

TABLE IV
UNIFIED FIVE-SEED CO-MITIGATION BASELINES

Method	Clean	PGD-10	Core min
Ours, next epoch	83.09 $\pm$ 0.57	46.72 $\pm$ 1.05	18.14 $\pm$ 0.58
Ours, immediate-switch	83.06 $\pm$ 0.65	47.05 $\pm$ 0.74	18.74 $\pm$ 0.63
FastAdv+	81.96 $\pm$ 0.60	47.09 $\pm$ 0.71	27.27 $\pm$ 3.53
GradAlign	79.54 $\pm$ 0.86	48.31 $\pm$ 0.81	45.71 $\pm$ 0.11

between clean and random-point input gradients, and a second-order graph only for the random branch. It otherwise uses batch size 128 and the shared SGD, momentum, weight decay, and triangular learning-rate schedule.

Table IV reports measured core parameter-update time. FastAdv+’s separately logged online PGD probes add 3.14 ± 0.00 minutes, giving 30.41 ± 3.53 measured minutes including those probes. Immediate-replay’s snapshot save-restore operations were not timed, so 18.68 minutes is not a complete end-to-end measurement. GradAlign achieves the highest mean PGD-10 accuracy among these 30-epoch methods; the proposed policy occupies a lower-robustness, lower-measured-core-cost point.

F. Strong Test-Set Evaluation

Table V reports full-test clean and PGD evaluation on all 10,000 CIFAR-10 test images. Table VI gives a supplementary AutoAttack-standard evaluation on the first 1,000 examples in the unshuffled official test-set order. AA-1k uses the ℓ_∞ threat model at $\epsilon = 8/255$ and the standard attack suite comprising APGD-CE, APGD-T, FAB-T, and Square. The immediate-switch, immediate-replay, next-epoch, FastAdv+, and GradAlign checkpoints use seed 101; the FGSM-RS and PGD-10 training references use seed 17. In each run, *best_robust.pt* is selected by the highest recorded validation PGD-10 accuracy, before test evaluation. This is a representative checkpoint comparison rather than a cross-seed aggregate. Validation PGD-10 selects checkpoints but does not trigger the online monitor. The immediate-switch seed-101 checkpoint was fixed before any result in Tables V–VI was observed; no alternative seed or checkpoint was tested for replacement. The 46.93% PGD-10 entry in Table V is from this single pre-fixed seed-101 checkpoint. Table II’s $47.05 \pm 0.74\%$ is instead the mean and sample standard deviation across five intervention seeds; the two summaries need not match.

For the primary immediate-switch method, PGD-50 and five-restart PGD-20 give 45.52% and 45.66%. Their agreement, and the analogous agreement for immediate-replay and the training references, does not reveal an obvious weak-iterative-attack failure. On the same fixed 1,000 images, the full standard AutoAttack suite gives 42.30% for immediate-switch, 42.60% for immediate-replay, 41.30% for next-epoch escalation, 43.00% for ordinary FGSM-RS, 48.70% for PGD-10 training, 42.40% for FastAdv+, and 44.40% for GradAlign. The two controlled baselines use their seed-101 checkpoints selected by the same validation rule. AA-1k is a subset estimate, so it complements the full-test PGD results without ruling out gradient masking.

TABLE V
FULL-TEST STRONG CIFAR-10 ACCURACY (%)

Method	Clean	PGD-10	PGD-50	PGD $\times$ 5
Immediate-switch (ours)	82.95	46.93	45.52	45.66
Immediate-replay (ablation)	82.97	47.20	45.46	45.65
Next epoch	82.57	46.38	44.75	44.91
Fast FGSM-RS	82.48	47.77	46.24	46.36
PGD-10 training	82.47	51.76	50.77	50.84

TABLE VI
AUTOATTACK-STANDARD ACCURACY ON THE FIXED FIRST 1,000
CIFAR-10 TEST IMAGES (%), ℓ_∞ , $\epsilon = 8/255$

Method	AA-1k
Immediate-switch (ours)	42.30
Immediate-replay (ablation)	42.60
Next epoch	41.30
Fast FGSM-RS	43.00
PGD-10 training	48.70
FastAdv+	42.40
GradAlign	44.40

V. DISCUSSION, COST, AND LIMITATIONS

A. Causal and Temporal Interpretation

The detector supplies a warning, PGD-2 restores robustness, and immediate-switch converts the warning into timely action so that PGD-2 can begin before CO. Immediate-switch matches immediate-replay’s 5/5 pre-collapse interventions and differs by only +0.22 PGD-10 points, with a paired interval crossing zero. Restoring the epoch snapshot is therefore not required for the observed timing benefit.

Table VII reports every matched timing event. High lead time is desirable for safety, but a warning far ahead of collapse does not imply precise prediction of collapse timing; the monitor acts as a safety trigger upon observing geometry degradation. In particular, the 1568- and 1680-update leads in seeds 23 and 202 should not be interpreted as calibrated time-to-CO forecasts. The saved-trace audit in the Appendix also checks whether the score remains above threshold after these early warnings.

B. Engineering Cost and Reproducibility

The attack-start gradient constructs the FGSM example. The endpoint input gradient is obtained from the same parameter-backward pass, without an additional forward or backward pass. The monitor retains it only until its cosine is reduced to a batch scalar, then stores scalar window history. It does not call a validation attack or an extra model evaluation.

The primary immediate-switch response does not require a snapshot. Its 18.74-minute measurement is still core training time, not complete end-to-end runtime. The immediate-replay control has a separate systems cost: its implementation holds one epoch-start copy of model, optimizer, and AMP-scaler state in device memory, plus Python, NumPy, PyTorch CPU/CUDA, and loader-generator RNG state. It does not write this immediate-replay snapshot to disk. Reported core training time does not include snapshot save-restore overhead. It does include abandoned FGSM parameter-update

TABLE VII
EVENT-LEVEL TIMING IN OPTIMIZER UPDATES

Seed	Warn→CO	Immediate-switch	Deferred	Deferred before CO
17	40	0	0	yes
23	1568	0	212	yes
42	20	0	232	no
101	60	0	292	no
202	1680	0	52	yes

time and the replayed PGD-2 epoch. The experiment did not instrument snapshot-copy latency or peak memory, so the current measurements must not be read as complete end-to-end latency or memory results.

Every detector choice is fixed before its associated held-out evaluation: 20-update windows, five-window median history, three-window minimum history, the event definition, and the 60-update near horizon. Fixed-switch controls ignore detector callbacks. Retrospective PGD labels never trigger the proposed monitor. Seed-level warning, collapse, PGD-start, accuracy, and timing records remain in the saved experiment CSV files; paired intervals use 100,000 resamples.

C. Scope and Limitations

The architecture result is partial unchanged-threshold transfer: ResNet-18 detects 4/5 events but only 2/5 near events. CIFAR-100 is not zero-shot or plug-and-play; transfer requires six dataset-specific development trajectories to refit the detector operating point using its original event-level lexicographic rule. Only one of its three collapse-prone development runs meets the CO definition, so the event-level calibration has limited statistical stability. As a supplementary check, a window-level audit contains one positive next-window label among 2,877 eligible windows. One alarm episode occurs across three held-out non-event runs; its unnecessary PGD-2 cost is unquantified. Event samples for ResNet-18, CIFAR-100, and the 16/255 stress test remain small, as reflected by wide Wilson intervals. Transfer evidence is still concentrated in CIFAR datasets and the ResNet family; ImageNet and substantially different architectures remain outside the current scope.

The primary immediate-switch checkpoint is included in full-test PGD-50 and multi-restart PGD, plus the full standard AutoAttack suite on a fixed 1,000-image subset, but not full-test AutoAttack. The subset is identical across compared checkpoints and includes APGD-CE, APGD-T, FAB-T, and Square. Agreement among PGD and this supplementary evaluation reduces, but does not eliminate, concern about gradient masking. The cosine score has empirical and mechanistic support but is neither a theoretical certificate nor a necessary or sufficient condition for CO. The 16/255 recovery results use immediate-replay; immediate-switch has not been tested in that stress setting. Immediate-replay also has snapshot memory, storage-policy, and engineering overhead not characterized for larger models. Immediate-switch avoids this snapshot and has both five-seed intervention and pre-fixed seed-101 strong evaluations in Tables V–VI.

GradAlign has higher mean PGD-10 robustness than the proposed policy, which uses lower measured core training time and provides an explicit event log, but the choice between them depends on whether the deployment prioritizes peak robustness, measured training cost, or auditable failure detection.

VI. CONCLUSION

We presented a gradient-consistency safety trigger for catastrophic overfitting and tested whether its warning arrives soon enough to change training. Seven held-out CIFAR-10/PreActResNet-18 events are detected within 60 updates. Transfer is partial with an unchanged threshold on ResNet-18 and requires a separate event-level calibration rule on CIFAR-100. In a randomized-FGSM 16/255 stress test, all three events are detected; the tested immediate-replay response raises terminal PGD-10 from zero to $26.48 \pm 0.83\%$.

For the pre-fixed seed-101 checkpoint of the primary immediate-switch response, full-test PGD-50 and five-restart PGD-20 accuracy are 45.52% and 45.66%, and AutoAttack-standard on the fixed first 1,000 test images gives 42.30%.

Across five seeds, the detector supplies the warning, PGD-2 restores robustness, and immediate-switch determines whether recovery starts in time. Immediate-switch and immediate-replay precede CO in 5/5 cases; rollback has no reliable final-accuracy advantage. In the controlled baselines, GradAlign obtains higher robustness and conditional escalation uses less measured core training time. The method is an empirical early-warning and conditional-computation mechanism, without a universal prevention claim, calibrated time-to-collapse output, or formal robustness guarantee.

APPENDIX

A. Immediate-Switch Paired Intervals

Table VIII reports all requested paired bootstrap comparisons for the five-seed immediate-switch experiment. Accuracy differences are percentage points; “before CO” is a difference in proportions. Each interval uses 100,000 paired resamples of seeds 17, 23, 42, 101, and 202.

TABLE VIII
IMMEDIATE-SWITCH MINUS THE MATCHED RESPONSE: MEAN DIFFERENCE [95% CI]

Metric	Immediate-replay	Next epoch
Before-CO proportion	0.00[0.00, 0.00]	+0.40[0.00, +0.80]
Warn→start (updates)	0.0[0.0, 0.0]	−157.6[−252.0, −56.8]
Clean (points)	−0.07[−0.16, 0.00]	−0.02[−0.14, +0.08]
PGD-10 (points)	+0.22[−0.22, +0.66]	+0.33[+0.08, +0.66]
Core time (min)	+0.06[−0.18, +0.30]	+0.59[+0.47, +0.71]

B. Offline Sensitivity Analysis

Below we perform offline sensitivity analysis on saved held-out traces, without retraining or modifying main-paper frozen parameters. The main protocol remains $\tau = 0.1663298875$, a 60-update near horizon, and the original CO definition. The threshold scan is descriptive and is not used to replace the frozen value. It uses the seven held-out

CIFAR-10 event trajectories and three separately reported randomized-FGSM non-event trajectories.

1) *Threshold Sensitivity*: Table IX recomputes warnings from saved cosine traces. Event recall, 60-update near recall, and non-event alarms are unchanged from 0.10 through 0.24. The median first-warning lead changes from 60 to 40 updates at thresholds of 0.20 and above. Event and false-alarm counts are stable around the frozen threshold; no post-hoc retuning is applied. Here “median first-warning lead” uses the first active warning anywhere before each event. Table I instead reports “median near lead”: the earliest active warning restricted to the final 60 updates. At the frozen threshold these deliberately different summaries are 60 and 40 updates, respectively.

TABLE IX

OFFLINE THRESHOLD SENSITIVITY ON SAVED TRACES. LEAD IS THE FIRST-WARNING LEAD; TABLE I REPORTS NEAR LEAD.

	τ	Recall	Near	Non-event alarms	Median first lead
0.1663298875 (frozen)	0.10	7/7	7/7	0	60
	0.12	7/7	7/7	0	60
	0.14	7/7	7/7	0	60
	0.16	7/7	7/7	0	60
	0.18	7/7	7/7	0	60
	0.20	7/7	7/7	0	40
	0.22	7/7	7/7	0	40
	0.24	7/7	7/7	0	40

2) *Warning-Horizon Sensitivity*: At the frozen threshold, Table X varies only the reporting horizon. Six of seven events have an active warning within 20 updates, and all seven are covered by 40 updates. The registered 60-update horizon is retained in the main paper.

TABLE X

OFFLINE WARNING-HORIZON SENSITIVITY

Horizon (updates)	Near events	Near recall	Non-event alarms
20	6/7	85.7%	0
40	7/7	100%	0
60 (frozen)	7/7	100%	0
80	7/7	100%	0
100	7/7	100%	0

3) *CO-Definition Sensitivity*: The original event requires a drop of at least 0.20 from the previous PGD-10 peak and accuracy at most 0.10. Table XI changes these labeling constants only after training. All seven trajectories remain events and are detected before the event under every variant. Near recall changes because stricter labels place the registered event later or earlier relative to saved warning windows; the main definition is unchanged.

4) *Long-Lead Signal Audit*: Table XII checks the two long-lead intervention traces. The score is not continuously above threshold: only four post-warning windows are active in each case. The first crossing is treated as a safety trigger after observed geometry degradation, not as a calibrated estimate of time to collapse.

TABLE XI

OFFLINE CO EVENT-DEFINITION SENSITIVITY

Definition	Drop	Ceiling	Events	Detected	Near
Relaxed	0.15	0.15	7	7/7	7/7
Original	0.20	0.10	7	7/7	7/7
Higher ceiling	0.20	0.15	7	7/7	7/7
Strict	0.25	0.05	7	7/7	5/7

TABLE XII

SIGNAL PERSISTENCE AFTER LONG-LEAD WARNINGS

Seed	Lead	Windows	Active	Longest active run
23	1568	80	4 (5.0%)	3
202	1680	86	4 (4.7%)	2

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